Hong Nguyen

Curriculum Vitae

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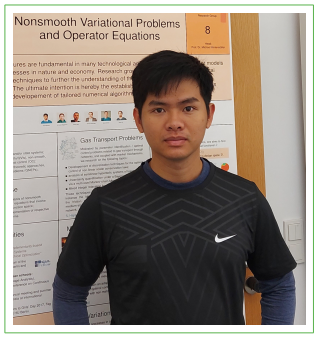
ResearchGate/Hong Quan Ba Nguyen



"The essence of mathematics lies in its freedom." - Georg Cantor
"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Education

2020–now **PhD student enrolled in Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis & Stochastics* (WIAS), Berlin, Germany.
In the Research Group Nonsmooth Variational Problems & Operator Equations, Head: Prof. Michael Hintermüller, within the European Innovative Training Network Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC), in the Project 11 Optimal Shape Design of Air Ducts in Combustion Engines.

2018–2019 **Master de Sciences, Technologie, Santé, Mathematiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics & Applications within ENS Rennes program (skipped Master 1).

2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University, Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis & Numerical Analysis.

2011–2014 **Elementary Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

Title *Staggered & well-balanced discretization of shallow-water equations* [pdf][slide]

Supervisor Prof. Nicolas Seguin

Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Research Interests

- **Mathematical Analysis** ▷ Partial Differential Equations (PDEs), Differential Geometry (DG).
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).
- **Mathematical Optimization** ▷ Optimal Control, Shape Optimization, Topology Optimization.
- **Fluid Mechanics** ▷ Fluid Dynamics, Computational Fluid Dynamics (CFDs).
- **Turbulence** ▷ Turbulence modeling.

Experience

Apr 2021–now **Software Development Engineer (SDE)**, *Math$_+$Tec GmbH*, Vienna, Austria.
Secondments for technical & scientific training at the industrial partner within ROMSOC project.

Mar–Jun 2019 **Master Internship**, IRMAR, Rennes, Bretagne, France.

Miscellaneous

2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students* (CBT), Vietnam.
Trained the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Publications

[1] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. "Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term". *Adv. Nonlinear Stud.* (2020). DOI https://doi.org/10.1515/ans-2020-2076. [pdf]

Awards & Scholarships

Mar 2020–Aug 2021 Marie Skłodowska-Curie Actions (MSCA) PhD Fellowship, *Horizon 2020 - Innovative Training Network - European Industrial Doctorate Programs*, Grant Agreement number: 765374 - Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC) [H2020-MSCA-ITN-EID/ROMSOC/ESR11].

2018–2019 Lebesgue Master Scholarship for *Master 2* program within ENS Rennes program.

2017 Scholarship of *Vietnam Institute for Advanced Studies in Mathematics* (VIASM).

2016 Special Prize ⇐ First Prize in Algebra + First Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2016).

2015 Second Prize in Algebra, Second Prize in Analysis, in *Vietnamese Mathematical Olympiad for College Students* (VMC 2015).

2012 Gold Medal ranked #2, in *Vietnamese Apr 30th Mathematical Olympiad for Central & Southern Vietnam*.

Computer Skills

GitHub GitHub/NQBH.

Basic R, Fireshape, Firedrake, FEniCS, Python 3, STAR-CCM+.

Intermediate git, C/C++, OpenFOAM.

Advanced UNIX/Linux [openSUSE, (K)Ubuntu], TEX/LATEX, MATLAB.

Languages

Vietnamese Mother tongue: *Literature-oriented*

English A *fluent speaker/reader* & an *advanced listener/writer*

Referees

[5] **Prof. Michael Hintermüller** *Primary PhD Supervisor*

[4] **Prof. Nicolas Seguin** *Master Supervisor*

[3] **Dr. Nguyen Anh Dao** *Bachelor Supervisor*

[2] **Dr. Tan Trung Nguyen** *Bachelor Teacher*

[1] **Dr. Manh Khang Dao** *Bachelor Teacher*

Updated to Apr 10, 2021.